

HW5

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```
suppressPackageStartupMessages(library(dplyr))
suppressPackageStartupMessages(library(lme4))
cult <- read.table('cult_data.dat', header = FALSE)
colnames(cult) <- c('field', 'cultivar', 'soil', 'weight')
cult <- cult %>%
  group_by(soil, cultivar) %>%
  mutate(group = cur_group_id()) %>%
  ungroup()
cult <- cult %>%
  mutate(across(c(field, cultivar, soil, group), as.factor))
```

1

```
fit <- lmer(weight ~ cultivar + soil + (1|field), data=cult)
summary(fit)
```

Linear mixed model fit by REML ['lmerMod']
Formula: weight ~ cultivar + soil + (1 | field)
Data: cult

REML criterion at convergence: 73.8

Scaled residuals:

Min	1Q	Median	3Q	Max
-1.66497	-0.65771	0.01962	0.41603	1.74666

Random effects:

Groups	Name	Variance	Std.Dev.
field	(Intercept)	1.213	1.101

```
Residual          1.163    1.078
Number of obs: 24, groups: field, 4
```

Fixed effects:

	Estimate	Std. Error	t value
(Intercept)	27.6458	0.7050	39.213
cultivarb	0.6333	0.4403	1.439
soildea	1.9875	0.5392	3.686
soilliv	5.3750	0.5392	9.968

Correlation of Fixed Effects:

	(Intr)	cltvr	soilde
cultivarb	-0.312		
soildea	-0.382	0.000	
soilliv	-0.382	0.000	0.500

The variance component of the random error is the 'Residual Variance' in the 'Random Effects' portion: 1.163

2

The between field variance component is the 'Field Variance' in the 'Random Effects' portion: 1.213

3

```
cell_means_mod <- lmer(weight ~ group - 1 + (1|field), data = cult)
summary(cell_means_mod)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: weight ~ group - 1 + (1 | field)
Data: cult
```

REML criterion at convergence: 68.5

Scaled residuals:

	Min	1Q	Median	3Q	Max
	-1.40580	-0.53790	0.06846	0.59451	1.84385

Random effects:

Groups	Name	Variance	Std.Dev.
field	(Intercept)	1.207	1.099
	Residual	1.196	1.094

Number of obs: 24, groups: field, 4

Fixed effects:

	Estimate	Std. Error	t value
group1	27.9000	0.7752	35.99
group2	28.0250	0.7752	36.15
group3	29.2500	0.7752	37.73
group4	30.6500	0.7752	39.54
group5	33.1500	0.7752	42.76
group6	33.5250	0.7752	43.25

Correlation of Fixed Effects:

	group1	group2	group3	group4	group5
group2	0.502				
group3	0.502	0.502			
group4	0.502	0.502	0.502		
group5	0.502	0.502	0.502	0.502	
group6	0.502	0.502	0.502	0.502	0.502

The estimate of the cell means is the estimates from the 'Fixed Effects' portion:

Group	Estimate
1	27.9
2	28.025
3	29.25
4	30.65
5	33.15
6	33.525